

Sports intelligence - Workbook

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QRU Sports Intelligence Lab(TM) Workbook

CHAPTER 1

Train Your Brain Like an Athlete Trains Their Body(TM)

1.1 A General Public Workbook for Building Math Confidence, Reasoning, and Real Understanding Through Sports

Welcome: Every Brain Can Train

There is hope for every learner.

A child who struggles with math is not "bad at math." They may need more practice, a different explanation, or a different strategy.

QRU Sports Intelligence Lab(TM) turns learning into a training experience. Just as athletes improve their bodies through repetition, feedback, adjustment, and reflection, Brain Athletes strengthen their minds through meaningful practice.

This workbook uses sports as a doorway into understanding.

The goal is not perfection.

The goal is growth.

1. The Big Idea: What Is Sports Intelligence?

Sports intelligence means using sports, strategy, practice, teamwork, and decision-making to strengthen the way we think and learn.

In QRU Sports Intelligence Lab(TM), mathematics becomes a mental sport.

Instead of learning multiplication as isolated memorization, learners connect math to:

- Sports
- Strategy
- Competition
- Practice
- Teamwork
- Decision-making

The learner becomes a:

CHAPTER 2

Brain Athlete

A Brain Athlete trains the mind through practice, reflection, and growth.

CHAPTER 3

Core QRU Principle

Children do not need more information.
They need stronger understanding systems.

CHAPTER 4

Why This Matters

Many children believe:

"I'm bad at math."

QRU teaches a stronger understanding:

"You may need more practice, different explanations, or a different strategy."

This preserves dignity while building capability.

Learning difficult things is possible.

2. Becoming a Brain Athlete

Every Brain Athlete begins as a rookie.

They enter the QRU Sports Intelligence Lab.

They train.

They practice.

They struggle.

They improve.

They discover that champions are not created by talent alone.

Champions are created through intentional practice.

CHAPTER 5

Brain Athlete Identity Statement

Read this out loud:

I can train my brain.

I can practice difficult things.

Mistakes are training reps.

Understanding creates confidence.

I can grow.

CHAPTER 6

Workbook Reflection

Answer in your own words.

6.1 1. What is one thing you want your brain to get stronger at?

6.2 2. What is one sport, game, or activity that helps you feel interested or excited?

6.3 3. When something is hard, what helps you keep trying?

3. The QRU Training Path

QRU means:

CHAPTER 7

Quest for Real Understanding(TM)

The QRU path follows this training sequence:

1. Question
 2. Understanding
 3. Practice
 4. Connection
 5. Application
 6. Reflection
 7. Growth
-

CHAPTER 8

What Each Step Means

8.1 Question

Start by asking what you are trying to understand.

8.2 Understanding

Do not only memorize. Learn what the idea means.

8.3 Practice

Repeat the skill in meaningful ways.

8.4 Connection

Connect the idea to something real, such as sports.

8.5 Application

Use the idea to solve a problem.

8.6 Reflection

Review what happened, like watching game film.

8.7 Growth

Use what you learned to improve next time.

CHAPTER 9

QRU Practice Page

Choose one math idea you are learning.

9.1 Question

What do I want to understand?

9.2 Understanding

What does this idea mean?

9.3 Practice

How can I practice it?

9.4 Connection

How can I connect it to sports?

9.5 Application

How can I use it in a problem?

9.6 Reflection

What did I notice?

9.7 Growth

What will I try next?

4. Multiplication as Many Groups

Combined

Multiplication is not just memorizing facts.

In QRU Sports Intelligence Lab(TM), multiplication means:

CHAPTER 10

Many groups combined.

CHAPTER 11

Vocabulary Decoder(TM)

11.1 Multiplication

Multi = many

plication = groups

11.2 Meaning

Multiplication means many groups combined.

CHAPTER 12

Example

If a quarterback completes 5 passes each quarter and there are 4 quarters:

$$5 \times 4 = 20 \text{ passes}$$

That means:

- 5 passes in each quarter
 - 4 quarters
 - 20 total passes
-

CHAPTER 13

Another Example

If a basketball player makes 8 free throws each practice and practices 5 times:

$$8 \times 5 = 40 \text{ free throws}$$

That means:

- 8 free throws each practice
 - 5 practices
 - 40 total free throws
-

CHAPTER 14

Practice: Many Groups Combined

Complete each sentence.

14.1 1. Multiplication means:

14.2 2. In 5×4 , there are:

- Number in each group:
 - Number of groups:
 - Total:
-

14.3 3. In 8×5 , there are:

- Number in each group:
 - Number of groups:
 - Total:
-

5. Sports Math in Action

Sports can help math feel connected to real life.

When athletes train, they repeat actions.

When Brain Athletes learn, they repeat thinking.

Athletes improve through:

- Repetition
- Feedback
- Adjustment
- Reflection

Learning follows a similar process.

CHAPTER 15

Football Training Problem

A quarterback completes 5 passes each quarter.
There are 4 quarters.

15.1 Step 1: Identify the groups

How many passes each quarter?

15.2 Step 2: Identify the number of groups

How many quarters?

15.3 Step 3: Multiply

$$5 \times 4 =$$

15.4 Step 4: Explain the meaning

Write a sentence explaining what the answer means.

CHAPTER 16

Basketball Training Problem

A player makes 8 free throws each practice.
The player practices 5 times.

16.1 Step 1: Identify the groups

How many free throws each practice?

16.2 Step 2: Identify the number of groups

How many practices?

16.3 Step 3: Multiply

$8 \times 5 =$

16.4 Step 4: Explain the meaning

Write a sentence explaining what the answer means.

6. Mistakes Are Training Reps

A mistake does not mean you cannot learn.

In QRU Sports Intelligence Lab(TM):

CHAPTER 17

Mistakes = Training Reps

Athletes do not improve by avoiding practice.

They improve by practicing, receiving feedback, adjusting, and

trying again.

Brain Athletes do the same.

CHAPTER 18

Analogy Library

| QRU Analogy | Meaning |

|---|---|

| Brain = Muscle | The brain strengthens through meaningful practice |

| Practice = Training | Learning takes repetition |

| Mistakes = Training Reps | Mistakes can help guide improvement |

| Knowledge = Equipment | Knowledge gives you tools |

| Understanding = Game Strategy | Understanding helps you know what to do |

| Reflection = Game Film Review | Reflection helps you improve next time |

CHAPTER 19

Reflection: A Mistake That Helped Me

Think of a time you made a mistake while learning.

19.1 1. What happened?

19.2 2. What did you learn?

19.3 3. What could you try differently next time?

19.4 4. How was that mistake like a training rep?

7. Strategy, Patterns, and Decisions

Sports involve strategy.

Learning also involves strategy.

When Brain Athletes understand patterns, they can make stronger decisions.

CHAPTER 20

Advanced Question

How does understanding patterns improve decisions?

Write your answer.

CHAPTER 21

Strategy Builder

Choose one multiplication problem you understand.

Problem:

21.1 1. What are the groups?

21.2 2. How many groups are there?

21.3 3. What pattern do you notice?

21.4 4. How can this pattern help you solve the problem?

8. Reflection: Game Film for the Brain

In sports, athletes may review what happened so they can improve.

In learning, reflection works like game film review.

You ask:

- What happened?
- What did I understand?
- Where did I get stuck?
- What strategy helped?

- What will I try next?
-

CHAPTER 22

Brain Game Film Review

Choose one problem you solved in this workbook.

Problem:

22.1 1. What did I do first?

22.2 2. What helped me understand?

22.3 3. Where did I feel stuck?

22.4 4. What strategy did I use?

22.5 5. What will I do next time?

9. Create Your Own Sports Problem

Now you become the problem designer.

Use sports, groups, and multiplication.

Remember:

Multiplication means many groups combined.

CHAPTER 23

Design Your Own Sports Problem

23.1 1. Choose a sport situation

Example idea: practice, quarters, free throws, passes, training reps, or another sports situation.

Write your situation:

23.2 2. Choose the number in each group

How many happen each time?

23.3 3. Choose the number of groups

How many times does it happen?

23.4 4. Write the multiplication sentence

$\times =$

23.5 5. Explain the answer

The answer means:

23.6 6. Explain your strategy

I solved it by:

10. Quiz and Review

This quiz includes recall, application, reasoning, and reflection questions.

CHAPTER 24

Part A: Recall Questions

24.1 1. What does QRU stand for?

- A. Quick Recall Unit
- B. Quest for Real Understanding(TM)
- C. Question Review Upgrade

Answer:

24.2 2. Multiplication means:

- A. Many groups combined
- B. Guessing quickly
- C. Racing against others

Answer:

24.3 3. In QRU Sports Intelligence Lab(TM), the learner becomes a:

- A. Brain Athlete
- B. Scorekeeper only
- C. Memorization machine

Answer:

CHAPTER 25

Part B: Application Questions

25.1 4. A quarterback completes 5 passes each quarter. There are 4 quarters. How many passes total?

Work:

Answer:

25.2 5. A basketball player makes 8 free throws each practice. The player practices 5 times. How many free

throws total?

Work:

Answer:

CHAPTER 26

Part C: Reasoning Questions

26.1 6. Why is multiplication more than memorization?

26.2 7. How can sports help someone understand math?

26.3 8. Why are mistakes called training reps?

CHAPTER 27

Part D: Reflection Questions

27.1 9. What is one thing you understand better after this workbook?

27.2 10. What is one strategy you can use next time learning feels difficult?

11. Growth Plan

The goal is not perfection.

The goal is helping the learner discover:

I can learn difficult things.

CHAPTER 28

My Brain Athlete Training Plan

28.1 Skill I want to improve:

28.2 How I will practice:

28.3 Who can coach, encourage, or guide me?

28.4 What I will do when I make a mistake:

28.5 How I will reflect:

28.6 What growth will look like:

12. Memory Sentences(TM)

Read these often.

Champions train their bodies.

Brain Athletes train their minds.

Understanding creates confidence.

Every brain can train.

Answer Key

CHAPTER 29

Practice: Many Groups Combined

29.1 1. Multiplication means:

Many groups combined.

29.2 2. In 5×4 :

- Number in each group: 5
- Number of groups: 4

- Total: 20

29.3 3. In 8×5 :

- Number in each group: 8
 - Number of groups: 5
 - Total: 40
-

CHAPTER 30

Football Training Problem

- Passes each quarter: 5
 - Number of quarters: 4
 - $5 \times 4 = 20$
 - Meaning: The quarterback completed 20 total passes.
-

CHAPTER 31

Basketball Training Problem

- Free throws each practice: 8
 - Number of practices: 5
 - $8 \times 5 = 40$
 - Meaning: The player made 40 total free throws.
-

Quiz Answer Key

31.1 1. What does QRU stand for?

B. Quest for Real Understanding(TM)

31.2 2. Multiplication means:

A. Many groups combined

31.3 3. In QRU Sports Intelligence Lab(TM), the learner becomes a:

A. Brain Athlete

31.4 4. Quarterback problem:

$5 \times 4 = 20$ passes

31.5 5. Basketball problem:

$8 \times 5 = 40$ free throws

31.6 6. Why is multiplication more than memorization?

Because multiplication means many groups combined, and understanding the meaning helps build confidence and reasoning.

31.7 7. How can sports help someone understand math?

Sports can connect math to practice, patterns, strategy, teamwork, and real-world problem solving.

31.8 8. Why are mistakes called training reps?

Because mistakes can help guide feedback, adjustment, reflection, and improvement.

Closing Reflection

A Brain Athlete does not need to know everything at the beginning.

A Brain Athlete begins with a question, practices with purpose, reflects with honesty, and grows with confidence.

Math is not only about correct answers.

It is also about understanding, reasoning, persistence, communication, and the belief that difficult things can be learned.

Continue your understanding at QRU:

